Optimization in Statistics

From Likelihood Geometry to Modern Algorithms

StatHarbor Analytics

Course Outline

Why Optimization Sits at the Heart of Statistics

Maximum Likelihood and Estimating Equations

Local Geometry of Objective Functions

Gradient Descent

Newton's Method and Fisher Scoring

EM Algorithm

Convexity

Why Optimization Sits at the Heart of Statistics

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- Optimization is therefore not peripheral to statistics: it is the computational machinery that turns a model into an estimator.

A Unifying View of Estimation

Many familiar estimators are optimization solutions

$$\text{OLS:} \quad \hat{\beta} = \arg\min_{\beta} \| Y - X\beta \|_2^2,$$

$$\text{MLE:} \quad \hat{\theta} = \arg\max_{\theta} \ell(\theta),$$

$$\text{Ridge:} \quad \hat{\beta} = \arg\min_{\beta} \left\{ \| Y - X\beta \|_2^2 + \lambda \|\beta\|_2^2 \right\},$$

$$\text{MAP:} \quad \hat{\theta} = \arg\max_{\theta} \{ \log p(y \mid \theta) + \log p(\theta) \}.$$

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- The statistical question defines the objective.
- The optimization algorithm determines how we compute the solution.

Objective Function, Estimator, Algorithm

Three distinct layers

1. **Objective:** what criterion defines "best fit"?
2. **Estimator:** what population quantity is targeted?
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Important distinction

The **objective function** defines the estimator. The **algorithm** is only a route to that estimator.

Optimization vs. Inference

Optimization

Finds a parameter value:

$$\hat{\theta}$$

that minimizes or maximizes an objective.

Inference

Quantifies uncertainty about that estimate:

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- Optimization answers: *where is the estimate?*
- Inference answers: *how precise is it?*

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Central message

Optimization is the engine of estimation, but statistics still asks whether the resulting estimator is identifiable, stable, and scientifically meaningful.

Maximum Likelihood and Estimating Equations

Maximum Likelihood Estimation

Suppose $Y_1, \ldots, Y_n$ are independent with density $f(y; \theta)$. The log-likelihood is

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- Likelihood converts probabilistic modeling into a numerical objective.
- The logarithm turns products into sums and makes derivatives tractable.

Why We Maximize the Log-Likelihood

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Taking logs:

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- Sums are algebraically easier than products.
- Numerical underflow is reduced.
- Derivatives decompose observation by observation.

The Score Function

The first derivative of the log-likelihood is the **score**:

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$$s(\hat{\theta}) = 0.$$

This is the multivariate analogue of locating stationary points by setting the derivative equal to zero.

MLE as an Estimating Equation

MLE is more than calculus. It can be viewed as solving a sample moment condition:

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Big picture

The score equation is an **estimating equation**. This places MLE in the same broad family as:

- method of moments,
- generalized estimating equations,
- generalized method of moments.

Worked Example: Normal Mean with Known Variance

Suppose

$$Y_i \sim \mathcal{N}(\mu, \sigma^2), \qquad \sigma^2 \text{ known.}$$

Then

$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i - \mu)^2.$$

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$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i - \mu)^2.$$

Maximizing $\ell(\mu)$ is equivalent to minimizing

$$\sum_{i=1}^{n}(Y_i - \mu)^2.$$

Normal Mean: Solving the Score Equation

Differentiate:

$$\frac{d\ell(\mu)}{d\mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(Y_i - \mu).$$

Set equal to zero:

$$\sum_{i=1}^{n}(Y_i - \hat{\mu}) = 0.$$

Therefore,

$$\hat{\mu} = \bar{Y}.$$

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Interpretation

The MLE balances positive and negative deviations exactly. The sample mean is the point where the score vanishes.

Second Derivative and Curvature

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Lesson

The first derivative identifies candidate optima. The second derivative tells us about local curvature and whether the point is a maximum, minimum, or saddle.

The Hessian Matrix

In multiple dimensions, curvature is summarized by the Hessian:

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Multivariate second-order condition

For maximization, curvature must bend downward in every direction.

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Statistical meaning

Sharper curvature means the likelihood peak is narrower. Narrower peaks correspond to more precise estimators.

Curvature and Asymptotic Variance

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$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\big(0, \mathcal{I}(\theta_0)^{-1}\big).$$

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Deep connection

Optimization geometry determines statistical precision.

Local Geometry of Objective Functions

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For a smooth scalar objective $f : \mathbb{R}^p \to \mathbb{R}$,

$$f(x) \approx f(x_k) + \nabla f(x_k)^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top H(x_k)(x - x_k).$$

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- Gradient = slope.
- Hessian = curvature.

First-Order Approximation

If we ignore curvature, the local model becomes

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This linear approximation says:

- f increases fastest in the direction of $\nabla f(x_k)$,
- f decreases fastest in the direction of $-\nabla f(x_k)$.

Why the Negative Gradient is the Steepest Descent Direction

Consider a unit vector d with $\|d\| = 1$. The directional derivative is

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Conclusion

The negative gradient is the direction of steepest local decrease.

Level Sets and Orthogonality

Level sets are contours of constant objective value:

$$\{x : f(x) = c\}.$$

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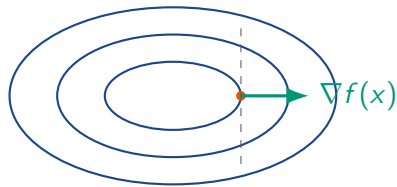
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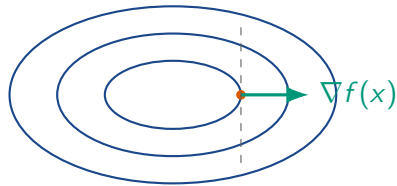
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contours of equal objective value

Optimization moves across contours; geometry determines how easy that movement is.

Conditioning

For quadratic objectives,

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- $\kappa \approx 1$: nearly spherical surface, easy optimization.
- Large κ: elongated valley, slow progress, zig-zagging.

Why Curvature Matters Numerically

Well-conditioned

Contours almost circular. Gradient steps point efficiently toward the optimum.

Ill-conditioned

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Key numerical insight

Statistical estimation problems can be theoretically well-defined yet computationally difficult because of poor curvature.

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$$x^{(k+1)} = x^{(k)} - \eta_k \nabla f(x^{(k)}).$$

- Move against the local slope.
- Recompute the gradient.
- Repeat until approximate stationarity.

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- Uses only first-order information.
- Easy to implement in high dimensions.
- Widely applicable beyond statistics, including machine learning and signal processing.

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But there is a price

Because it ignores curvature, gradient descent can be slow when the surface is ill-conditioned.

Role of the Step Size

The step size η_k determines how far we move in the descent direction.

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For an L-smooth convex objective, constant step sizes satisfying

$$0 < \eta < \frac{2}{L}$$

guarantee descent stability.

Descent Lemma Intuition

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Plugging in the gradient step $y = x - \eta \nabla f(x)$ gives

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So if $0 < \eta < 2/L$, the objective decreases.

Worked Example: Least Squares

Consider

$$f(\beta) = \frac{1}{2}\|Y - X\beta\|_2^2.$$

Then

$$\nabla f(\beta) = -X^\top (Y - X\beta) = X^\top X\beta - X^\top Y.$$

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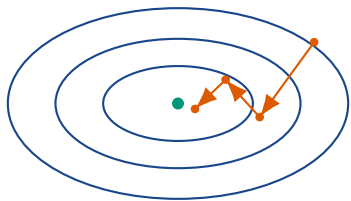
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OLS has a closed-form solution, but this iterative view is foundational for large-scale estimation.

Geometric Interpretation of Gradient Descent



typical zig-zag path in an elongated valley

Convergence Perspective

For smooth convex problems:

- gradient descent converges globally,
- but often only at a linear or sublinear rate,
- and the rate depends strongly on curvature.

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Practical lesson

Gradient descent is simple and general, but simplicity does not imply speed.

Newton's Method and Fisher Scoring

Newton's Method: The Basic Idea

Instead of using only slope, Newton's method uses a second-order local model:

$$f(x) \approx f(x_k) + \nabla f(x_k)^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top H(x_k)(x - x_k).$$

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Minimizing this quadratic approximation yields a curvature-adjusted update.

Deriving the Newton Update

Differentiate the quadratic approximation with respect to x:

$$\nabla f(x_k) + H(x_k)(x - x_k) = 0.$$

Solving for x gives

$$x = x_k - H(x_k)^{-1} \nabla f(x_k).$$

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Therefore Newton's method updates as

$$x^{(k+1)} = x^{(k)} - H(x^{(k)})^{-1} \nabla f(x^{(k)}).$$

Why Newton Can Be Much Faster

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Near a well-behaved optimum, Newton's method has **quadratic convergence**: the number of correct digits can roughly double each iteration.

Limitations of Newton's Method

- Requires computing the Hessian.
- Requires solving a linear system or inverting the Hessian.
- Can be unstable far from the optimum.
- If the Hessian is indefinite, the Newton step may not be a descent step.

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Important

Newton is locally brilliant but not automatically globally safe.

Maximization Version for Likelihood

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Because the Hessian of a concave log-likelihood is negative definite near the maximum, this update moves toward the optimizer.

Fisher Scoring

Fisher scoring replaces the observed Hessian with the expected information:

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This is often more stable because expected curvature is smoother than the observed second derivative.

Why Fisher Scoring Matters in Statistics

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- It connects asymptotic variance and optimization in a very direct way.
- It uses a statistically meaningful curvature matrix rather than a purely numerical one.

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Interpretation

Fisher scoring is a Newton-like method adapted to the structure of likelihood-based inference.

Worked Example: Logistic Regression

For binary $Y_i \in \{0, 1\}$,

$$\Pr(Y_i = 1 \mid x_i) = \pi_i = \frac{\exp(x_i^\top \beta)}{1 + \exp(x_i^\top \beta)}.$$

The log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \Big[Y_i x_i^\top \beta - \log(1 + \exp(x_i^\top \beta)) \Big].$$

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No closed-form maximizer exists, so iterative optimization is required.

Logistic Regression: Score and Hessian

The score is

$$s(\beta) = X^\top (Y - \pi),$$

and the Hessian is

$$H(\beta) = -X^\top W X,$$

where W is diagonal with entries $\pi_i(1 - \pi_i)$.

Logistic Regression: Score and Hessian

The score is

$$s(\beta) = X^\top (Y - \pi),$$

and the Hessian is

$$H(\beta) = -X^\top W X,$$

where W is diagonal with entries $\pi_i(1 - \pi_i)$.
Therefore the Newton/Fisher scoring step becomes

$$\beta^{(k+1)} = \beta^{(k)} + (X^\top W X)^{-1} X^\top (Y - \pi).$$

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This is the core of iteratively reweighted least squares.

EM Algorithm

The Latent Variable Problem

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The integral inside the logarithm often makes direct maximization difficult.

The Strategy of EM

EM alternates between:

- computing expectations with respect to the latent variables,
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EM alternates between:

- computing expectations with respect to the latent variables,
- maximizing a simpler surrogate objective.

It converts an intractable direct maximization problem into a sequence of easier problems.

The E-Step

Given current parameter value $\theta^{(t)}$, define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Z|X,\theta^{(t)}}\big[\log p(X, Z; \theta)\big].$$

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This replaces the unobserved complete-data log-likelihood by its conditional expectation.

The M-Step

Update the parameter by maximizing the surrogate:

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The M-step is usually easier than maximizing the observed-data likelihood directly.

Why EM Works: Jensen's Inequality

Write

$$\log p(X;\theta) = \log \int q(Z) \frac{p(X,Z;\theta)}{q(Z)}\, dZ.$$

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If we choose $q(Z) = p(Z \mid X, \theta^{(t)})$, the bound is tight at $\theta^{(t)}$.

Monotonicity of EM

Each EM iteration satisfies

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Each EM iteration satisfies

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- EM never decreases the observed-data log-likelihood.
- It converges to a stationary point, typically a local maximum.

Important limitation

EM is monotone, but it can be slow and can converge to local rather than global optima.

Example Contexts for EM

EM is widely used in:

- Gaussian mixture models,
- missing data problems,
- hidden Markov models,
- latent class models.

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- Gaussian mixture models,
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Conceptual interpretation

EM climbs the likelihood surface indirectly by alternating between imputation-like averaging and parameter optimization.

Convexity

Definition of Convexity

A function $f : \mathbb{R}^p \to \mathbb{R}$ is convex if for all x, y and $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y).$$

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Geometrically, the graph of a convex function lies below every secant line joining two points on the graph.

Why Convexity Matters

Convexity transforms optimization from a potentially fragile search problem into a much more reliable one.

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Optimization guarantee

Convexity removes the problem of spurious local minima.

First-Order Characterization

If f is differentiable and convex, then for all x, y,

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x).$$

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$$f(y) \geq f(x) + \nabla f(x)^\top (y - x).$$

This means the tangent plane is a global under-estimator.
If $\nabla f(x^*) = 0$, then

$$f(y) \geq f(x^*) \qquad \text{for all } y,$$

so x^* is a global minimizer.

Second-Order Characterization

If f is twice differentiable, convexity is equivalent to

$$H(x) \succeq 0 \qquad \text{for all } x,$$

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Curvature now guarantees not only descent behavior but uniqueness of the solution.

Convexity in Statistics

Many important statistical objectives are convex:

- ordinary least squares,
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Many important statistical objectives are convex:

- ordinary least squares,
- ridge regression,
- logistic regression negative log-likelihood,
- lasso objective.

Consequences

When the objective is convex, algorithm design becomes easier and interpretation becomes cleaner because every local optimum is already the solution we want.

Regularization and Penalized Estimation

Why Regularization?

In high-dimensional or unstable problems, fitting the data too closely can inflate variance.

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Regularization modifies the optimization problem:

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The penalty discourages complexity in exchange for improved stability.

Bias–Variance Tradeoff

- Unpenalized estimators can have low bias but large variance.
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- Penalized estimators intentionally introduce bias to reduce variance.

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Regularization is useful when variance is the dominant problem.

Ridge Regression

Ridge solves

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \left\{ \frac{1}{2n} \| Y - X\beta \|_2^2 + \lambda \| \beta \|_2^2 \right\}.$$

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The ℓ_1 penalty is not differentiable at zero.
That non-smoothness is not a nuisance by accident; it is the source of sparsity.

Geometry of Ridge and Lasso

Ridge

$$\|\beta\|_2^2 \leq c$$

gives circular or spherical constraint regions.

Lasso

$$\|\beta\|_1 \leq c$$

gives diamond-shaped or polyhedral constraint regions.

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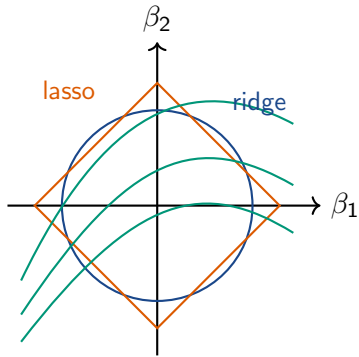
Lasso

$$\|\beta\|_1 \leq c$$

gives diamond-shaped or polyhedral constraint regions.

Because lasso has sharp corners on the coordinate axes, the optimizer is more likely to land exactly on a coordinate axis, forcing some coefficients to zero.

Geometric Picture



Subgradient Idea for Lasso

For $g(\beta) = |\beta|$, the derivative fails at 0, but the subgradient exists:

$$\partial|\beta| = \begin{cases} \{1\}, & \beta > 0, \\ [-1, 1], & \beta = 0, \\ \{-1\}, & \beta < 0. \end{cases}$$

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This generalized derivative framework lets us optimize non-smooth objectives rigorously.

Stochastic and Large-Scale Optimization

Why Full Gradient Methods Can Be Expensive

For large datasets,

$$\nabla f(\theta) = \sum_{i=1}^{n} \nabla f_i(\theta)$$

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can be costly to compute at every iteration.
If n is massive, repeated full passes through the data become the computational bottleneck.

Stochastic Gradient Descent

SGD replaces the full gradient by a noisy single-observation or mini-batch approximation:

$$\theta^{(k+1)} = \theta^{(k)} - \eta_k \nabla f_{i_k}(\theta^{(k)}).$$

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- cheaper iterations,
- noisier updates,
- often much faster in large-scale problems.

Mini-Batch Gradient Descent

A compromise between full gradient and SGD is the mini-batch update:

$$\theta^{(k+1)} = \theta^{(k)} - \eta_k \frac{1}{|B_k|} \sum_{i \in B_k} \nabla f_i(\theta^{(k)}).$$

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- less variable than single-sample SGD,
- cheaper than full-gradient updates,
- well suited to modern parallel computing.

Learning Rate Schedules

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Common choices include:

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A constant step size can speed early learning, while decay helps stabilize late iterations.

Statistical Interpretation of SGD Noise

The stochastic gradient is unbiased in many settings:

$$\mathbb{E}[\nabla f_{i_k}(\theta)] = \nabla f(\theta).$$

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But it has variance:

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Tradeoff

Stochastic methods reduce computation per iteration at the cost of noisier optimization trajectories.

Quasi-Newton and Second-Order Approximations

Why Quasi-Newton Methods?

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Newton's method is powerful but can be expensive because it needs the Hessian. Quasi-Newton methods aim to keep the speed advantages of second-order methods without explicitly recomputing the full Hessian each time.

Basic Quasi-Newton Idea

Instead of H_k, build an approximation B_k to the Hessian, or B_k^{-1} to its inverse, from gradient differences across iterations.

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This injects curvature information while avoiding exact second derivatives.

BFGS in Words

BFGS is a popular quasi-Newton method that:

- starts with an initial matrix approximation,
- updates it using displacement and gradient changes,
- preserves positive definiteness under suitable conditions.

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- starts with an initial matrix approximation,
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It is often highly effective in smooth medium-scale problems.

When Quasi-Newton is Attractive

- Smooth objective
- Dimension not too massive
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Practical place in the toolkit

Quasi-Newton methods are often the "smart default" for smooth deterministic optimization.

Constrained Optimization

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Many statistical problems are not unconstrained. We may need

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$$\min_x f(x) \qquad \text{subject to} \qquad g_j(x) \leq 0,\ j = 1, \ldots, m.$$

Examples:

- probabilities must lie in $[0, 1]$,
- covariance matrices must be positive semidefinite,
- mixing weights must sum to one.

The Lagrangian

For a single inequality constraint $g(x) \leq 0$, define

$$\mathcal{L}(x, \lambda) = f(x) + \lambda g(x), \qquad \lambda \geq 0.$$

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The multiplier λ measures how strongly the constraint influences the optimum.

KKT Conditions

Under appropriate regularity conditions, a constrained optimum satisfies:

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0,$$

$$g(x^*) \leq 0, \qquad \lambda^* \geq 0,$$

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The last condition is called **complementary slackness**.

Interpretation of Complementary Slackness

$$\lambda^* g(x^*) = 0$$

means:

- either the constraint is inactive, so $g(x^*) < 0$ and $\lambda^* = 0$,
- or the constraint binds exactly, so $g(x^*) = 0$ and λ^* may be positive.

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- or the constraint binds exactly, so $g(x^*) = 0$ and λ^* may be positive.

This tells us which constraints actually determine the optimum.

Proximal Algorithms for Non-Smooth Problems

When Smooth Calculus Breaks

Not all optimization problems are differentiable. A common form is

$$\min_x \big\{ f(x) + g(x) \big\},$$

where

- f is smooth,
- g is convex but non-smooth.

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- f is smooth,
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Lasso is the canonical example with $g(x) = \lambda \|x\|_1$.

The Proximal Operator

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This is a compromise between:

- staying close to v,
- respecting the structure encouraged by g.

Proximal Gradient Descent

For

$$\min_x \{f(x) + g(x)\},$$

the proximal gradient update is

$$x^{(k+1)} = \text{prox}_{\eta g}\left(x^{(k)} - \eta \nabla f(x^{(k)})\right).$$

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It is the natural extension of gradient descent to objectives with a non-smooth part.

Soft Thresholding for the Lasso

When $g(x) = \lambda \|x\|_1$, the proximal operator becomes soft thresholding:

$$S_{\eta\lambda}(v_j) = \text{sign}(v_j)\big(|v_j| - \eta\lambda\big)_+.$$

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This shrinks coefficients toward zero and sets sufficiently small ones exactly to zero.

Why this matters

Sparsity is created by the optimization operator itself, not added after estimation.

Optimization Across Statistical Models

Linear Regression

Ordinary least squares solves

$$\hat{\beta} = \arg\min_{\beta} \|Y - X\beta\|_2^2.$$

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$$\hat{\beta} = \arg\min_{\beta} \|Y - X\beta\|_2^2.$$

Because the objective is quadratic and convex, the optimizer is explicit:

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$

when $X^\top X$ is invertible.

Logistic Regression

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Consequences:

- no closed-form solution,
- iterative methods are required,
- Newton or Fisher scoring are often preferred.

Generalized Linear Models

GLMs unify many models through

$$g(\mathbb{E}[Y \mid X]) = X\beta.$$

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Optimization typically proceeds through:

- score equations,
- curvature-based updates,
- iteratively reweighted least squares.

Mixture Models and Latent Structures

Mixtures, hidden states, and missing-data problems often produce complicated likelihood surfaces with:

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EM becomes a natural algorithmic framework.

Neural Networks

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Even here, the core ideas remain: gradients, curvature, regularization, and generalization.

Optimization Challenges

Local Minima and Global Minima

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In non-convex problems, not every stationary point is equally good.
A local minimum may satisfy

$$f(x^*) \leq f(x) \quad \text{for nearby } x,$$

without being the best point globally.

Saddle Points

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$$\nabla f(x^*) = 0$$

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Saddle points are common in high-dimensional non-convex optimization.

Ill-Conditioning Revisited

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Preconditioning and curvature-aware methods are often used to address this.

Initialization Matters

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- convergence speed,
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This is why multiple starts are common in mixture modeling and clustering.

Algorithmic Comparison

Comparing Major Optimization Strategies

Method	Uses	Strength	Weakness
Gradient descent	Gradient	Simple, scalable	Can be slow
Newton	Gradient + Hessian	Very fast locally	Expensive, unstable far away
Fisher scoring	Score + expected curvature	Natural for likelihoods	Still second-order
SGD / mini-batch	Noisy gradients	Massive-scale data	Variable trajectory
Proximal gradient	Gradient + prox map	Handles non-smooth penalties	Needs proximal structure
EM	Conditional expectations + maximization	Elegant for latent variables	Usually only local, often slow

How to Choose an Algorithm

Choice depends on:

- the shape of the objective,
- whether the problem is smooth or non-smooth,
- whether data are large-scale,
- whether latent variables are present,
- whether exact second derivatives are available and worthwhile.

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Practical rule

The right algorithm is determined by the objective geometry and computational regime, not by fashion.

Statistical Perspective

Algorithm Does Not Define the Estimand

Two algorithms may optimize the same objective and therefore target the same estimator.

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- Newton and gradient descent can both target the MLE,
- coordinate descent and proximal gradient can both target the lasso solution.

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For example:

- Newton and gradient descent can both target the MLE,
- coordinate descent and proximal gradient can both target the lasso solution.

What changes is computation, not the statistical target.

Better Optimization Does Not Automatically Mean Better Science

Driving training loss lower can:

- improve fit,
- worsen overfitting,
- reduce interpretability,
- amplify instability in weakly identified settings.

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Statistics asks not only whether we can optimize, but whether the optimized answer is scientifically trustworthy.

Optimization, Regularization, and Generalization

Modern statistical learning often relies on the joint action of:

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The optimization problem is only one piece of a broader inferential workflow.

Key Takeaways

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1. Many statistical estimators are solutions to optimization problems.
2. The gradient encodes local slope; the Hessian encodes curvature.
3. MLE is both an optimization problem and an estimating-equation problem.
4. Convexity provides global guarantees; non-convexity introduces multiple local solutions.
5. Regularization changes optimization geometry to improve stability.
6. Different algorithms serve different objective structures.

Final Synthesis

Optimization in statistics is not just numerical computation

It links:

- model assumptions,
- geometry of objective functions,
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Core lesson

To understand an estimator deeply, you must understand the optimization problem that defines it.

Learn it. Prove it. Ship it.

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References & Further Reading

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